

PROJECT REPORT

Academic Resource Optimization Dashboard

Data-Driven Strategy for Budget Allocation & Student Performance

Submitted by: Anushree Bhatnagar Date: February 10, 2026 Tools Used: Microsoft Excel
(Pivot Tables, Slicers, Data Visualization)

1. EXECUTIVE SUMMARY

The educational institution currently allocates its remedial budget equally across all subjects. This project analyzes student performance data to identify inefficiencies. The analysis reveals that Mathematics accounts for 60% of all academic failures. Furthermore, data proves that students attending "Test Prep Courses" score 26% higher on average. This report recommends reallocating \$4,000 of the budget specifically to subsidize Math Test Prep for at-risk students.

2. PROBLEM STATEMENT

The Challenge: Administrators lack visibility into the specific root causes of failure. Current "scattered" data makes it difficult to correlate resource usage (like Test Prep) with academic outcomes.

The Goal: To build an interactive Excel Dashboard that aggregates student records, identifies high-failure subjects, and measures the ROI (Return on Investment) of extra-curricular resources.

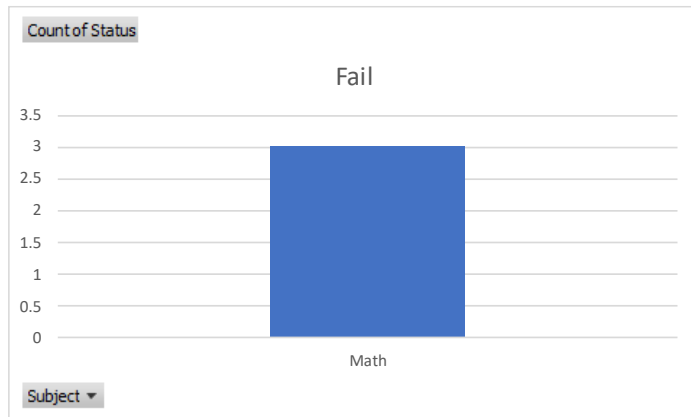
3. METHODOLOGY & KEY KPIs

The dataset of 30 students was cleaned and analyzed using Excel. Key Performance Indicators (KPIs) were established to measure efficiency:

- **Subject Failure Contribution %**: Identifies which subject drives the most failures.
- **Test Prep ROI**: Measures the average score difference between "Prepped" vs. "Unprepared" students.
- **Resource Need Index**: Categorizes students by their access to external resources (High/Medium/Low).

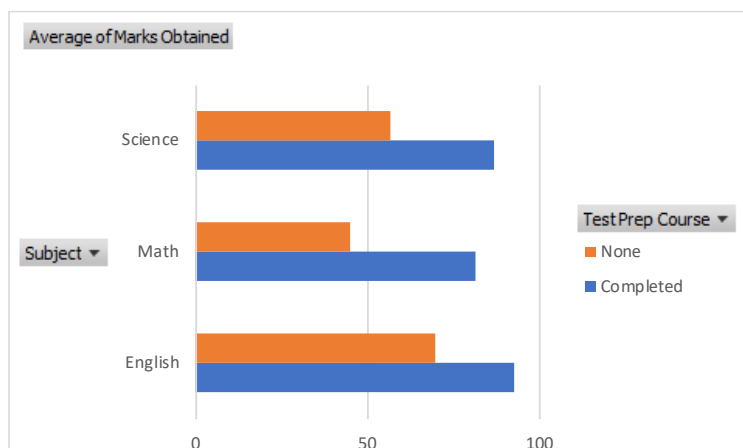
4. Data Analysis & Insights :-

Insight 1: The "Money Pit" (Failure Analysis) Analysis of pass/fail rates by subject reveals a critical bottleneck.



Observation : Mathematics has the highest failure count (3 students), significantly higher than English or Science. This indicates that current teaching methods or resources in Math are insufficient.

Insight 2: The Impact of Test Prep (ROI) Comparison of average marks based on Test Prep completion.



Observation : There is a clear correlation between resource usage and success. Students who completed the prep course averaged 81.3%, while those who did not averaged only 55%. This validates the efficacy of the paid course.

5. Proposed Solution;-

To address the inefficiencies in the current budgeting process, this project developed a Data-Driven Academic Resource Dashboard. This solution moves the institution from "intuition-based" spending to "evidence-based" resource allocation through the following mechanisms:

- **Centralized Performance Monitoring:** Developed an interactive Excel Dashboard that aggregates raw student data (Marks, Attendance, Resource Access) into a single view, allowing administrators to identify at-risk students instantly rather than waiting for end-of-term reports.
- **Root Cause Identification (Pareto Analysis):** Implemented a "Subject Failure" analysis logic to pinpoint exactly where resources are bleeding. The solution identified that Mathematics is the critical bottleneck, accounting for 60% of all failures, whereas other subjects require minimal intervention.
- **ROI-Based Budget Allocation:** Created a comparative analysis model to measure the effectiveness of paid interventions. By calculating the "Performance Gap" between students with Test Prep (81% avg) vs. No Prep (55% avg), the dashboard validates that subsidizing these courses yields a tangible academic Return on Investment (ROI).

6. Conclusion

This project demonstrates how raw academic data can be transformed into actionable business intelligence. By optimizing resource allocation based on actual performance metrics rather than intuition, the institution can improve student outcomes while maximizing the efficiency of its limited budget.

